Pedestrian dynamics in confined environments

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Human mobility is a famously scale-free phenomenon, but only where there is room to be free. Tracking pedestrians in a Barcelona campus, we find that the persistence of movement remains intact, while its heavy-tailed Lévy-like signature dissolves, a reminder that universal laws need space to emerge.

Well into the 21^{st} century, it comes as no surprise that most new technologies track our location, even when we are not aware of it. Despite the obvious cost to privacy, this opens the door to very interesting and rich social studies, among which the study of human mobility is one of the most prominent. Seen from far enough away, human mobility follows surprisingly regular patterns. Data from bank notes, mobile phones and social-media activity provide clear evidence of it since they trace trajectories whose jumps follow a power-law distribution, a hallmark of scale-free random walks (Lévy flights), and whose spread over time is superdiffusive [1, 2]. However, there is a hidden assumption in this picture: all of these "laws" were measured at the scale of cities, countries or the planet, where we can consider that a traveller is free to make a jump of any length. What happens to these laws when the space is limited?

A bounded playground.

To address this question, five groups of students recorded GPS trajectories using a mobile tracking application (Wikiloc) while walking through a campus district in Barcelona (Zona Universitària), covering an area of a few hundred metres over the course of about an hour. The resulting trajectories are shown in Fig. 1, traversed at an average speed of 1.32 ± 0.68 m/s (mean $\pm$ standard deviation), consistent with the typical pace of pedestrians but with a broad spread that reflects the frequent stops and changes of pace of the walkers. Raw coordinates, in terms of latitude and longitude, were projected onto a local metric frame and cleaned of logging gaps and spurious speed spikes before analysis. Let's now see what the data tell us about the persistence and scale-free nature of human mobility in this confined space.

Reading the tracks.

Taken together, three classical mobility descriptors tell a coherent and revealing story (Fig. 2).

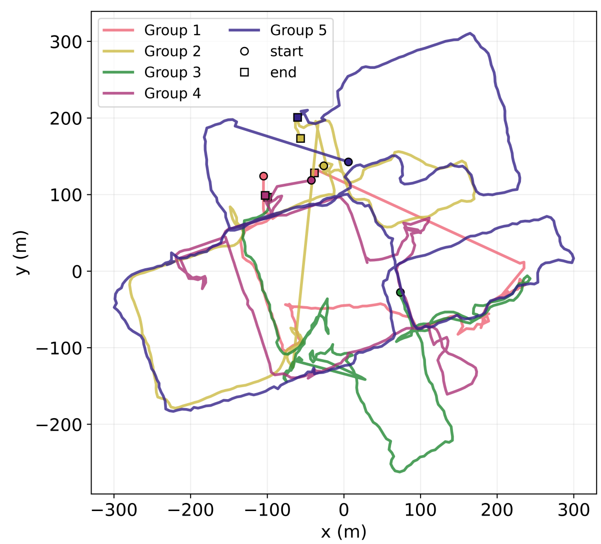

FIG. 1: Pedestrian tracks of the five groups in Zona Universitària, projected to a local metric frame (origin at the common centroid).

In the first panel, we analysed the time-averaged Mean-Squared Displacement, $\mathrm{MSD}(\tau) = \langle|\mathbf{r}(t+\tau) - \mathbf{r}(t)|^2\rangle$, where the time lag τ is the elapsed time between two points along a trajectory, averaged over all such pairs. Commonly believed to scale as $\mathrm{MSD}(\tau) \sim \tau^{\alpha}$ [2], in this case it does not: at short lags (4–30 s, red band) it grows almost ballistically, $\alpha \simeq 1.82$, while at long lags (100–550 s, blue band) it relaxes to a near-diffusive regime, $\alpha \simeq 1.20$. This crossover reflects the directional persistence of walking: over short times pedestrians keep their heading, almost walking in straight lines, whereas over long times accumulated turns decorrelate the direction and the motion gradually becomes more diffusive.

The second panel makes this persistence more explicit. We collected the turning angles θ between consecutive steps and fitted their distribution to a von Mises law, $p(\theta) \propto e^{\kappa \cos\theta}$, the circular analogue of a Gaussian, where κ measures the persistence of the motion ($\kappa = 0$ corresponding to a memoryless random walk). The resulting distribution presents a sharp peak around $\theta = 0$, with a concentration $\kappa \simeq 3.4$, confirming that the walkers prefer to keep going the way they were already going, as suggested by the near-ballistic short-lag MSD. Still, even if they are rare, large turns do occur and accumulate over time, which gradually erases the directional memory and explains the decay of α at long lags.

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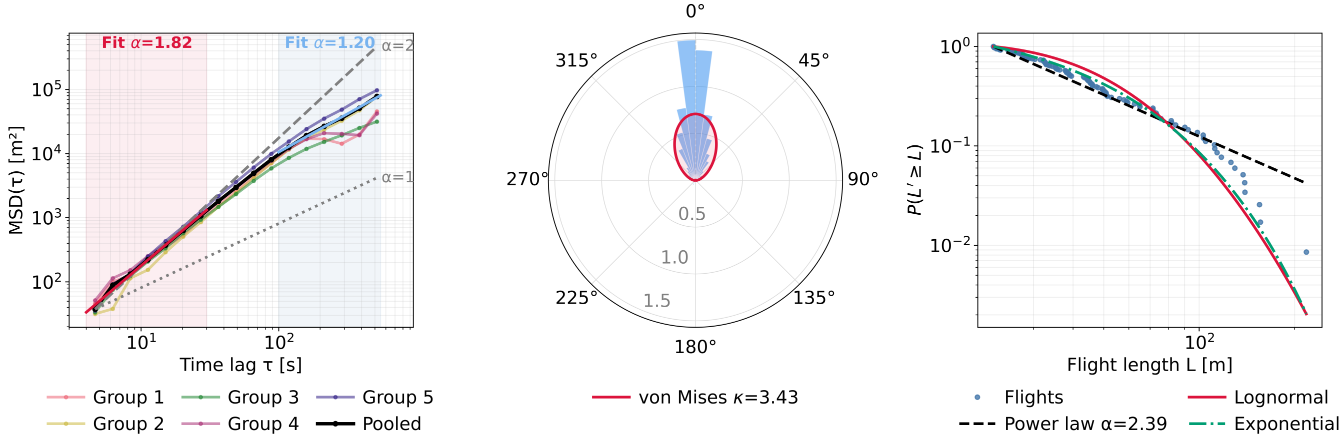

FIG. 2: Mobility signatures under confinement. (a) Time-averaged MSD, obtained by binning point pairs by their real time lag; short-lag motion is near-ballistic ($\alpha \simeq 1.82$, red band) and long-lag motion near-diffusive ($\alpha \simeq 1.20$, blue band); dotted/dashed guides show slopes 1 and 2. (b) Turning-angle distribution with a von Mises fit, $\kappa \simeq 3.4$. (c) Complementary cumulative distribution of flight lengths — straight runs broken whenever the heading turned by more than 30° — with power-law, log-normal and exponential fits to the tail estimated by maximum likelihood; the three are statistically indistinguishable.

Lastly, the third descriptor is where the story turns. We extracted “flights” from the trajectories [3], defined as straight runs ended whenever the heading deviated by more than 30°, and examined the tail of their length distribution (panel c). In the figure, we can observe that the tail falls off sharply beyond $L \sim 150\,\mathrm{m}$: of nearly 400 flights, only a few exceed this length. This can be understood as a direct fingerprint of a confined geometry: no jump can be longer than the space allows. As a result, the heavy tail that defines Lévy flights at larger scales never develops. In fact, if we perform a likelihood-ratio test to compare the power-law, log-normal and exponential fits to the tail, we find that they are statistically indistinguishable ($|z| < 2$), confirming that the scale-free signature has dissolved into the hard wall of a finite space.

Conclusions.

Overall, our small experiment carries a large lesson: confinement does not degrade movement uniformly but selectively. While the tendency to keep walking in the same direction survived almost untouched, the rare long flights that give mobility its scale-free character disappeared. In other words, while persistence knows nothing of the size of the world, the scale-free tail is precisely what a boundary forbids, which is consistent with the fact that scale-free laws are by definition laws without a characteristic scale. As a consequence, the moment a spatial contraint is introduced, the picture changes.

Zooming out, this has a clear implication for the study of human behaviour: we tend to think that the laws that are presented in the literature are a property of the walker alone, but they are not. The same individual can be a Lévy flyer across a city and a merely persistent walker across a courtyard. As a consequence, the “universal laws” drawn from large mobility datasets may describe the spaces people move through as much as the people themselves. Change the space, and the law changes with it. However, it is true that our experiment is very limited: only five short tracks, recorded in about an hour by walkers who knew they were being tracked and who set out to explore rather than to follow their usual routes. Still, we find this simplicity can be seen as encouraging: if a single afternoon on a campus is enough to keep a universal law from appearing, then the scale at which we observe a complex system is not a minor detail: universal laws need space to emerge.

Data and code availability.

The GPS trajectories and analysis code used in this work are openly available in the GitHub repository:
`https://github.com/abby-lr-ub/Pedestrian-dynamics-in-confined-environments.git`

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